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Electronic apparatus for power transfer using mechanical power transfer scheme, wireless power transfer method, and wireless power transfer system

Abstract

According to one embodiment, an electronic apparatus capable of transferring power wirelessly to a power receiver apparatus, apparatus includes: a signal generation circuitry configured to generate a first signal for wireless power transfer; an antenna configured to radiate an electromagnetic wave on a basis of the first signal; an actuator capable of mechanically changing an orientation of the antenna; and a controlling circuitry configured to control a transmission power of the first signal according to the orientation of the antenna.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7936095	12/2010	Kawabata	N/A	N/A
9800060	12/2016	Shimokawa	N/A	H02J 50/90
11804875	12/2022	Murata et al.	N/A	N/A
12107437	12/2023	Hamana et al.	N/A	N/A
12191678	12/2024	Taniguchi et al.	N/A	N/A
2006/0273756	12/2005	Bowling	320/107	B60L 53/68
2010/0217353	12/2009	Forsell	607/61	H02J 50/80
2012/0188917	12/2011	Knox	370/277	H01Q 1/2225
2015/0001959	12/2014	Shimokawa	307/104	H02J 50/10
2021/0184728	12/2020	Kang	N/A	H04B 5/266
2021/0328460	12/2020	Kuroda	N/A	G05D 1/0088

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
114256994	12/2021	CN	N/A
H8-130840	12/1995	JP	N/A
2007-267110	12/2006	JP	N/A
2008-92704	12/2007	JP	N/A
2010-258846	12/2009	JP	N/A
2013-132134	12/2012	JP	N/A
6512722	12/2018	JP	N/A
2020-14154	12/2019	JP	N/A
2021-125983	12/2020	JP	N/A
2021-151061	12/2020	JP	N/A

OTHER PUBLICATIONS

Kazuki Arai et al., “A Tile-based 8×8 Triangular Grid Array Beamformer for 5.7 GHz Microwave Power Transmission,” 2021 IEEE Radio and Wireless Symposium, pp. 101-104, DOI: 10.1109/RWS50353.2021.9360351 (2021). cited by applicant
Japanese Patent Office, Office Action in JP Application No. 2022-119879 (Jun. 3, 2025). cited by applicant

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is based upon and claims the benefit of priority from the prior Japanese Patent Application No. 2022-119879, filed on Jul. 27, 2022, the entire contents of which are incorporated herein by reference.

FIELD

(2) Embodiments herein related to an electronic apparatus, a wireless power transfer method, and a wireless power transfer system.

BACKGROUND

(3) There is a method of transferring power by microwaves with maximum efficiency to a receiver target placed at a given location through beamforming with an array antenna. Known beamforming methods include beamforming by digital control and beamforming by analog control.

(4) However, to perform beamforming, a transmitter circuit and a receiver circuit that can handle a large number of antenna elements are necessary, which increases component costs and implementation costs. One method of addressing this problem is a mechanical power transfer scheme in which a motor or the like is used to control the elevation angle and the azimuth angle (rotation angle) of a high-gain antenna. By adopting this method, a beamforming device is no longer necessary and costs are reduced.

(5) In the case of power transfer by microwaves, a situation in which another wireless system exists nearby is anticipated. In a mixed situation of a power transfer system and another wireless system, a problem occurs in that if the power transfer apparatus handles high power, coexistence with the other wireless system may be difficult with the mechanical power transfer scheme.

(6) That is, in the case of the electronic beamforming scheme described above, the maximum directivity is determined according to the directivity of individual antennas (e.g., patch antennas) forming an array antenna. For this reason, it is also possible to form a beam that suppresses radiation in an undesired direction (for example, a direction horizontal to the installation surface of the power transfer apparatus or a direction parallel to the ground). Consequently, high-power signal leakage to another wireless system on a neighboring frequency band can be reduced, even if the other wireless system is installed outside the power transfer area of the power transfer apparatus. Accordingly, coexistence and co-use with another wireless system is easy.

(7) On the other hand, in the case of the mechanical power transfer scheme, since the elevation angle of the antenna itself is changed to control the beam direction, the maximum directivity is always determined by the orientation of the antenna, independent of the elevation angle. Consequently, depending on the elevation angle, radiation in an undesired direction cannot be suppressed. For example, high-power leakage can occur if the main lobe is directed in the undesired direction, and even if the main lobe is not directed in the undesired direction, high-power leakage due to side lobes can still occur in the undesired direction. Therefore, coexistence with another wireless system that exists outside the power transfer area is difficult.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram illustrating an example of a microwave power transfer apparatus according to a first embodiment;

- (2) FIG. 2 is a diagram illustrating an installation example of the microwave power transfer apparatus in FIG. 1;
- (3) FIGS. 3A and 3B illustrate elevation angle-maximum output characteristics when electronic beamforming is performed;
- (4) FIGS. 4A and 4B illustrate elevation angle-maximum output characteristics when the mechanical power transfer scheme is employed;
- (5) FIGS. 5A and 5B illustrate an example of output control depending on the elevation angle;
- (6) FIG. 6 is a flowchart of an example of operations performed by the microwave power transfer apparatus according to an embodiment;
- (7) FIG. 7 is a block diagram illustrating an example of a microwave power transfer apparatus according to a second embodiment;
- (8) FIG. 8 is a block diagram illustrating an example of a microwave power transfer apparatus according to a third embodiment;
- (9) FIG. 9 is a block diagram illustrating an example of a microwave power transfer apparatus according to a fourth embodiment;
- (10) FIG. 10 is a block diagram illustrating an example of a microwave power transfer apparatus according to a fifth embodiment;
- (11) FIG. 11 is a block diagram illustrating an example of a microwave power transfer apparatus according to a sixth embodiment;
- (12) FIG. 12 is a block diagram illustrating an example of a microwave power transfer apparatus according to a seventh embodiment; and
- (13) FIG. 13 is a block diagram of a microwave power transfer apparatus according to an eighth embodiment.

DETAILED DESCRIPTION

(14) According to one embodiment, an electronic apparatus capable of transferring power wirelessly to a power receiver apparatus, apparatus includes: a signal generation circuitry configured to generate a first signal for wireless power transfer; an antenna configured to radiate an electromagnetic wave on a basis of the first signal; an actuator capable of mechanically changing an orientation of the antenna; and a controlling circuitry configured to control a transmission power of the first signal according to the orientation of the antenna.

(15) Hereinafter, embodiments of the present invention will be described with reference to the drawings.

First Embodiment

(16) FIG. 1 is a block diagram illustrating an example of a microwave power transfer apparatus **100** as an electronic apparatus or a wireless power transfer apparatus according to a first embodiment.

(17) FIG. 2 illustrates an installation example of the microwave power transfer apparatus **100** in FIG. 1. The microwave power transfer apparatus **100** is installed on a side (back) surface of a ceiling **160**. The microwave power transfer apparatus **100** is an apparatus that transfers power wirelessly to a power receiver apparatus **150** by using the mechanical power transfer scheme. The orientation of the antenna **101** can be changed in at least an elevation angle direction **D1** from among the elevation direction **D1** and an azimuth angle direction **D2**. Hereinafter, the direction opposite the ceiling **160** (the direction to the ground) may be referred to as downward in some cases. The installation location of the microwave power transfer apparatus **100** is not limited to the ceiling and may also be on a ground surface or any other location.

(18) As illustrated in FIG. 1, the microwave power transfer apparatus **100** is provided with an antenna **101**, a high-frequency device **103** (signal generator or signal generation circuitry), a mechanical elevation angle controller **102** (actuator), and a controller **104**.

(19) The high-frequency device **103** (signal generator) generates a signal (power transfer signal or first signal) for wireless power transfer to the power receiver apparatus **150**. The generated power transfer signal is supplied to the antenna **101**. In one example, the power transfer signal is a high-

frequency analog signal. The high-frequency device **103** generates the power transfer signal by amplifying a local signal produced by a local oscillator, for example. Before amplification, the local signal may be frequency-converted, and the frequency-converted signal may also be subject to bandwidth control using a filter.

(20) The mechanical elevation angle controller **102** (actuator) is a device which is connected to the antenna **101** and which can mechanically change the elevation angle from among and the elevation angle and the azimuth angle that define the orientation of the antenna **101**. The present embodiment deals with the case of changing the elevation angle as the orientation of the antenna **101**, but the case of changing the azimuth angle is dealt with in another embodiment.

(21) The controller **104** is a controlling circuitry including an orientation controller **104A** and a power controller **104B** (a power control circuit). The controller **104** may be configured by using a circuit such as an application-specific integrated circuit (ASIC) or a field-programmable gate array (FPGA), or by using a processor. Alternatively, some or all of these elements may be executed by a CPU that runs a program.

(22) The orientation controller **104A** is connected to the mechanical elevation angle controller **102** and changes the elevation angle of the antenna **101** by controlling the mechanical elevation angle controller **102**. The orientation controller **104A** controls the elevation angle of the antenna **101** by sending a command value specifying an elevation angle for the antenna **101** to the mechanical elevation angle controller **102**. The mechanical elevation angle controller **102** changes the elevation angle of the antenna **101** according to the command value. Information about the elevation angle to be set in the antenna **101** may also be set in advance. Alternatively, the orientation controller **104A** may acquire information about the elevation angle to be set in the antenna **101** from a user of the microwave power transfer apparatus **100** through an input interface (not illustrated) provided in the microwave power transfer apparatus **100**. Alternatively, information indicating the elevation angle to be set in the antenna **101** may be acquired from a user terminal operated by the user.

(23) The power controller **104B** is connected to the high-frequency device **103** and controls the transmission power (output power) of the power transfer signal generated by the high-frequency device **103** according to information about the elevation angle set in the antenna **101**. Control of the transmission power includes controlling the gain or attenuation of the high-frequency device **103**. The power controller **104B** controls the transmission power of the power transfer signal by sending a command value specifying the gain or attenuation, or alternatively the output power value or the like, to the high-frequency device **103**. In this way, by controlling the transmission power of the power transfer signal according to the elevation angle of the antenna **101**, the radiation pattern of electromagnetic waves radiated from the antenna **101** is controlled so as not to exceed an allowed value in an undesired direction (for example, the horizontal direction), irrespectively of the orientation of the antenna **101**. In one example, if the undesired direction is the horizontal direction, the maximum output value is 1/100 the maximum radiation level. This value may be set in the specifications. The undesired direction is the direction parallel to the installation surface of the microwave power transfer apparatus, for example.

(24) The antenna **101** is an antenna that has a radiation pattern having a specific directivity in the direction in which the antenna **101** faces. For example, maximum directivity is obtained in the direction that the antenna **101** faces. For example, when the antenna **101** is directed with an elevation angle of 0 degrees (downward), there is formed a radiation pattern having a directivity in which the maximum gain is downward. If the magnitude of the transmission power is the same, each radiation pattern generally has the same shape and size, irrespectively of the orientation of the antenna **101**. When the antenna **101** is directed with an elevation angle of 30 degrees (diagonally left downward), there is formed a radiation pattern having a directivity in which the maximum gain is diagonally left downward. The antenna **101** is connected to the high-frequency device **103**, receives the power transfer signal from the high-frequency device **103**, and radiates electromagnetic waves into space with the above-described directivity. The antenna **101** includes an

antenna element that radiates electromagnetic waves. The power receiver apparatus **150** can receive a transfer of power by receiving radiated electromagnetic waves. By controlling the elevation angle and the like of the antenna **101** with the mechanical elevation angle controller **102**, it is possible to transfer power to any power receiver apparatus, including the power receiver apparatus **150**.

(25) In this way, by controlling the transmission power of the power transfer signal according to the elevation angle of the antenna **101**, electromagnetic wave leakage exceeding an allowed value in an undesired direction can be prevented regardless of the antenna orientation, even in the case of a mechanical power transfer control (or elevation angle control) scheme in which the orientation of the antenna **101** and the orientation of maximum directivity exist in a 1-to-1 correspondence. Hereinafter, FIGS. **3** and **4** will be used to provide a more detailed description.

(26) FIG. **3A** illustrates maximum output characteristics **C1** for each elevation angle in the case of performing electronic beamforming with an array antenna that serves as a first comparative example. FIG. **3B** schematically illustrates a situation in which a beam is transmitted (a power transfer signal is transmitted) straight downward by electronic beamforming. A power transfer apparatus **1000** is installed on a ceiling **1001**.

(27) FIG. **4A** illustrates maximum output characteristics **C2** for each elevation angle in the case of performing power transfer according to another mechanical power transfer control (or elevation angle control) scheme that serves as a second comparative example. FIG. **4B** schematically illustrates a situation in which a beam is transmitted (a power transfer signal is transmitted) in the direction of approximately 305 degrees according to the power transfer control of the other mechanical power transfer control scheme.

(28) In the electronic beamforming scheme, the maximum directivity is determined according to the directivity of individual antennas (for example, patch antennas) forming an array antenna in the power transfer apparatus **1000**. Accordingly, by adjusting the directivity of the individual antennas (the weighting of the signals for the antennas), the desired elevation angle-output characteristics **C1**, like those illustrated in FIG. **3A** for example, are obtained easily. In other words, with electronic beamforming, it is also easy to point the maximum directivity in any direction to satisfy the maximum output characteristics **C1** illustrated in FIG. **3A**. By satisfying the maximum output characteristics **C1**, interference with another wireless system is mitigated, and coexistence with the other wireless system is easy.

(29) For example, in the case of transmitting a beam straight downward (0 degrees) from the power transfer apparatus **1000** installed on the ceiling **1001** as in FIG. **3B**, a radiation pattern **P1** is obtained. Also, in the case of transmitting a beam in the direction of approximately 314 degrees, the result satisfies the maximum output characteristics **C1** and a radiation pattern **P2** is obtained. Neither the main lobe nor the side lobes of the radiation patterns **P1** and **P2** exceed the maximum value (allowed value) indicated by the maximum output characteristics **C1**.

(30) With the maximum output characteristics **C1** in FIG. **3A**, the maximum output value is smallest in the horizontal direction of the installation surface of the power transfer apparatus **1000** or the direction parallel to the ground, which correspond to the undesired direction. The radiation pattern can be controlled so that no matter which direction a beam is transmitted in, neither the main lobe nor the side lobes have a maximum output value in the horizontal direction that exceeds the output value indicated by the maximum output characteristics **C1**. With this configuration, radiation in the horizontal direction can be suppressed. Accordingly, high-power signal leakage to another wireless system on a neighboring frequency band installed outside the power transfer area of the power transfer apparatus can be reduced, and therefore coexistence and co-use with the other wireless system is easy.

(31) On the other hand, FIG. **4A** illustrates the case of another mechanical power transfer control scheme that serves as a second comparative example, in which the maximum directivity is fixed (the maximum output value is the same for all directions). With the mechanical power transfer control scheme, since the elevation angle of the antenna itself is changed to control the beam

direction, the maximum directivity is fixed, irrespectively of the elevation angle. For this reason, the maximum output characteristics C2 illustrated in the diagram have a semicircular shape. Thus, since the directivity cannot be attenuated in the horizontal direction treated as the undesired direction, as the beam direction approaches the horizontal direction, high-power leakage due to at least one of the main lobe and the side lobes occurs in the horizontal direction without being attenuated. As a result, interference is imparted to another wireless system existing outside the power transfer area, making coexistence with the other wireless system difficult.

(32) For example, as in FIG. 4B, a radiation pattern P4 in the case of transmitting a beam in the direction of approximately 305 degrees and a radiation pattern P3 in the case of transmitting a beam straight downward (0 degrees) from a power transfer apparatus 2000 installed on a ceiling 2001 both have approximately the same maximum output power, and the shapes of the patterns are also substantially the same. Since directivity (or directivity attenuation) control in the horizontal direction is unavailable, a side lobe 51 (FIG. 4A) of the radiation pattern P4 has a power value in the horizontal direction that exceeds the allowed value.

(33) In contrast, with the mechanical power transfer control scheme according to the present embodiment, the controller 104 controls the gain, attenuation, or the like of the high-frequency device 103 on the basis of information about the elevation angle of the antenna 101. With this configuration, elevation angle-maximum output characteristics can be achieved such that the maximum output value in the undesired direction does not exceed the allowed value, irrespectively of the elevation angle (orientation) of the antenna 101.

(34) FIG. 5 is a diagram for explaining an example of gain or attenuation control performed by the controller 104.

(35) FIG. 5A schematically illustrates an example of gain or attenuation control performed by the controller 104. The gain or attenuation of the high-frequency device 103 is controlled (L1 to L6) to make the elevation angle-maximum output characteristics C2 (FIG. 4A) of the mechanical power transfer control in the second comparative example described above match or approach the elevation angle-maximum output characteristics C1 (FIG. 3B) of the electronic beamforming in the first comparative example. With this configuration, the elevation angle-maximum output characteristics C2 are modified. With the mechanical power transfer control in the second comparative example, the elevation angle-maximum output characteristics do not change depending on the elevation angle, but in the present embodiment, the gain or attenuation of the high-frequency device 103 is controlled according to the elevation angle such that the elevation angle-maximum output characteristics are the same or similar to those of the electronic beamforming in the first comparative example. According to the elevation angle-maximum output characteristics, for example, if the maximum output of the main lobe in the radiation pattern for any elevation angle direction is less than or equal to the maximum value of the maximum output characteristic in that elevation angle direction, the maximum output of the side lobes will also satisfy the elevation angle-maximum output characteristics.

(36) FIG. 5B illustrates a plot of a specific example of the control of the gain or attenuation. The horizontal axis is the elevation angle of the antenna 101, while the vertical axis is the attenuation. The attenuation is zero straight downward, and the attenuation increases as the elevation angle of the antenna 101 approaches the horizontal direction going both ways (−90 degrees (270 degrees), 90 degrees). That is, the maximum output when transferring power at each elevation angle is controlled to be the value of the maximum output characteristics C1.

(37) In this way, by controlling the attenuation according to the value of the elevation angle, electromagnetic leakage in the horizontal direction can also be reduced in the mechanical power transfer control scheme, similarly to electronic beamforming. That is, a reduction in electromagnetic leakage in the horizontal direction can be achieved in accordance with the actual usage environment.

(38) The plot in FIG. 5B is an example, and it is not strictly necessary to obey this plot. For

example, a null (or notch) portion of extremely low power gain may exist between the main lobe and the side lobes, or between side lobes, of the radiated electromagnetic waves. Accordingly, when this portion is pointed in the undesired direction (for example, the horizontal direction), a reduction in electromagnetic leakage in the horizontal direction may be attained in some cases, even if the attenuation is smaller compared to the case of obeying the plot in FIG. 58. That is, the plot (or elevation angle-maximum output characteristics) in FIG. 5B may be thought of as stipulating the maximum power value for each direction when the main lobe is pointed in that direction. For example, in the case of the horizontal direction, the maximum power value in the horizontal direction is stipulated for when the main lobe is pointed in the horizontal direction. Consequently, in cases in which the low portion described above is pointed in the horizontal direction and the main lobe is pointed in another direction, even if the main lobe is at the maximum value according to the plot, the low portion is marginally below the maximum output value in the horizontal direction. Consequently, in this case, it is also possible to increase the maximum output value of the main lobe over the case of obeying the plot. However, this assumes that the low portion does not exceed the maximum output value in the horizontal direction in the above plot (or elevation angle-maximum output characteristics). This configuration enables power transfer at high power, improving the power transfer efficiency.

(39) FIG. 6 is a flowchart of an example of operations performed by the microwave power transfer apparatus 100 according to the present embodiment. The operations correspond to an example of a wireless power transfer method according to the present embodiment. The controller 104 changes, via the actuator 102, the orientation of the antenna 101 relative to the power receiver apparatus 150 (S101). The controller 104 determines the transmission power of the signal for power transfer (first signal) according to the orientation of the antenna 101 (S102). The high-frequency device 103 generates the signal for power transfer on the basis of the determined transmission power (S103). The antenna 101 receives the generated signal and radiates electromagnetic waves into space. The maximum directivity of the radiation pattern is determined by the orientation of the antenna 101 (mechanical power transfer control scheme), but the size of the radiation pattern varies depending on the transmission power. Accordingly, the leakage of power exceeding the allowed value in the undesired direction due to the orientation of the antenna 101 can be prevented.

(40) As above, according to the present embodiment, by controlling the transmission power of a high-frequency signal according to the elevation angle of the antenna 101, the leakage of electromagnetic waves in an undesired direction (for example, the horizontal direction) can be reduced, even in the case of enacting a mechanical power transfer control. With this configuration, the influence on another wireless system existing in the horizontal direction can be reduced with a simple configuration. Thus, coexistence and co-use with another wireless system in the case of using the mechanical power transfer control scheme is easy.

Second Embodiment

(41) FIG. 7 is a block diagram illustrating an example of a microwave power transfer apparatus 200 as an electronic apparatus according to a second embodiment. The microwave power transfer apparatus 200 is provided with an antenna 201, a mechanical elevation angle controller 202 (actuator), a variable gain amplifier/variable attenuator 203, a high-frequency oscillator 204 (signal generator), and a controller 205. The microwave power transfer apparatus 200 is a specific implementation of the high-frequency device 103 of the microwave power transfer apparatus 100 in FIG. 1. The antenna 201, the mechanical elevation angle controller 202, and an orientation controller 205A in the controller 205 have the same functions as the antenna 101, the mechanical elevation angle controller 102, and the orientation controller 104A in the first embodiment, and therefore a description is omitted. Hereinafter, the differences from the first embodiment will be described mainly, whereas the description that is the same for the first and second embodiments will be omitted, as appropriate.

(42) The high-frequency oscillator 204 is an oscillator that generates a high-frequency signal on a

fixed frequency. The high-frequency oscillator **204** supplies the generated high-frequency signal to the variable gain amplifier/variable attenuator **203**.

(43) The variable gain amplifier/variable attenuator **203** includes at least one of a variable gain amplifier and a variable attenuator, or a set of a variable gain amplifier and a variable attenuator.

(44) The controller **205** includes an orientation controller **205A** and a power controller **205B**. The power controller **205B** controls the gain/attenuation of the variable gain amplifier/variable attenuator **203** according to information about the elevation angle set in the antenna **201**. The method of control is similar to the first embodiment.

(45) The variable gain amplifier/variable attenuator **203** amplifies the high-frequency signal supplied from the high-frequency oscillator **204** on the basis of a gain/attenuation set by the power controller **205B**. The amplified high-frequency signal is supplied to the antenna **201** as the power transfer signal, and electromagnetic waves are radiated from the antenna **201** by mechanical beamforming. The radiated electromagnetic waves are received by the power receiver apparatus **150** (see FIG. 2), thereby transferring power to the power receiver apparatus **150**.

(46) As above, according to the present embodiment, the gain/attenuation of the variable gain amplifier/variable attenuator **203** is controlled according to the elevation angle of the antenna **101**. With this configuration, the transmission power can be controlled with a simple configuration.

Third Embodiment

(47) FIG. 8 is a block diagram illustrating an example of a microwave power transfer apparatus **300** as an electronic apparatus according to a third embodiment.

(48) The microwave power transfer apparatus **300** is provided with an antenna **301**, a mechanical elevation angle controller **302** (actuator), a high-frequency device **303**, and a controller **304**. The antenna **301** and the mechanical elevation angle controller **302** have the same functions as the antenna **101** and the mechanical elevation angle controller **102** in the first embodiment, and therefore a description is omitted. Hereinafter, the differences from the first and second embodiments will be described mainly, whereas the description that is the same as the first and second embodiments will be omitted, as appropriate.

(49) The high-frequency device **303** is provided with a transmitter device **305**, a receive processor **306** (receive processing circuitry), and a switch **307**.

(50) The transmitter device **305** has functions similar to the high-frequency device **103** (see FIG. 1) of the first embodiment. Alternatively, the transmitter device **305** has functions similar to the variable gain amplifier/variable attenuator **203** and the high-frequency oscillator **204** in FIG. 7.

(51) The switch **307** includes a switch that selectively connects the antenna **301** to the receive processor **306** or the transmitter device **305**. The switch **307** receives instruction information specifying the connection destination from the controller **304**, and switches the connection destination in accordance with the instruction information.

(52) The receive processor **306** has a carrier sense function that detects whether another wireless system using the same or a neighboring frequency to the frequency used by the microwave power transfer apparatus **300** exists nearby. The receive processor **306** compares the receive power level of a signal (second signal) received through the antenna **301** to a threshold value. The receive processor **306** detects that another wireless system exists if the receive power level is equal to or greater than the threshold value, and detects that another wireless system does not exist if the receive power level is less than the threshold value. The threshold value for carrier sense can be set or switched by the controller **304**. The receive processor **306** sends carrier sense result information to the controller **304**.

(53) Note that in the case where an array antenna including multiple antenna elements for reception is provided, the receive processor **306** may also estimate a channel with the power receiver apparatus **150** by receiving beacon signals transmitted periodically from the power receiver apparatus **150**, the number of beacon signals being equal to the number of elements in the array antenna. In this case, the receive processor **306** may also estimate the direction in which the power

receiver apparatus **150** exists, on the basis of the result of the channel estimation. The receive processor **306** may also send direction information indicating the estimated result to the controller **304**.

(54) The controller **304** includes an orientation controller **304A** and a power controller **304B**.

(55) The orientation controller **304A** determines the elevation angle of the antenna **301** on the basis of the carrier sense result information. For example, the receive processor **306** is made to perform the carrier sense while varying the elevation angle of the antenna **301**, and the elevation angle of the antenna **301** is determined to be an angle at which the carrier sense result information does not detect another wireless system. Alternatively, the orientation controller **304A** acquires direction information from the receive processor **306**, determines the direction of the power receiver apparatus **150** on the basis of the direction information, and determines the elevation angle of the antenna **301** in accordance with the direction of the power receiver apparatus **150**. The orientation controller **304A** sends information indicating the determined elevation angle to the mechanical elevation angle controller **302**, causing the mechanical elevation angle controller **302** to set the elevation angle of the antenna **301**. Otherwise, information is sent to the mechanical elevation angle controller **302** on the basis of an elevation angle table set in advance by the user, causing the mechanical elevation angle controller **302** to set the elevation angle of the antenna **301**. An elevation angle setting pattern to be used when performing carrier sense while varying the elevation angle of the antenna **301** may also be stored in the elevation angle table.

(56) The power controller **304B** controls the transmission power of the power transfer signal according to a method similar to the first or second embodiment. In addition, the power controller **304B** determines the threshold value for carrier sense performed by the receive processor **306** according to the elevation angle determined by the orientation controller **304A**. In one example, the larger the elevation angle is, the larger the value of the threshold value is set. For instance, a first threshold value is determined for the case in which the elevation angle of the antenna points in a first direction, and a second threshold value greater than the first threshold value is determined for the case in which the elevation angle of the antenna points in a second direction closer to the horizontal direction than the first direction. In other words, in this case, the receive processor **306** uses the first threshold value when the elevation angle of the antenna points in a first direction, and uses the second threshold value greater than the first threshold value when the elevation angle of the antenna points in the second direction closer to the horizontal direction than the first direction. The power controller **304B** may also determine a threshold value corresponding to the elevation angle determined by the orientation controller **304A**, on the basis of a table in which elevation angles and threshold values are associated.

(57) The power controller **304B** controls power transfer to the power receiver apparatus **150** on the basis of the carrier sense result information from the receive processor **306**. For example, power transfer is initiated if, after the elevation angle of the antenna **301** is determined by the orientation controller **304A**, the carrier sense result information indicates that another wireless system using the same or a neighboring frequency does not exist nearby. Additionally, after power transfer is initiated, the power transfer is suspended if the carrier sense result information indicates that another wireless system using the same or a neighboring frequency exists nearby.

(58) According to the present embodiment, when using carrier sense to detect the presence or absence of another wireless system on the same or a neighboring frequency, setting a threshold value dependent on the elevation angle of the antenna makes it possible to perform carrier sense with an appropriate (not excessive) receive sensitivity according to the elevation angle. For example, the transmission power from the power transfer apparatus falls as the elevation angle diverges further from 0 degrees (see FIG. 5(B)), but in this case, since the transmission power exerts little influence on the other wireless system, the other wireless system should not be detected by carrier sense that is more sensitive than necessary. In the present embodiment, the threshold value for carrier sense is set according to the elevation angle, and therefore a suspension of power

transfer or a change of the elevation angle of the antenna due to unnecessary detection of the other wireless system can be mitigated. Thus, an improvement in the time efficiency of power transfer is possible.

Fourth Embodiment

(59) FIG. 9 is a diagram illustrating a configuration example of a microwave power transfer apparatus **400** as an electronic apparatus according to a fourth embodiment.

(60) The microwave power transfer apparatus **400** is provided with an antenna **401**, a mechanical elevation angle controller **402**, and a mechanical rotation angle controller **403**. The antenna **401** has the same function as the antenna **101** in the first embodiment. The high-frequency device and controller (orientation controller/power controller) according to any of the first to third embodiments are provided inside the housing of at least one of the mechanical elevation angle controller **402** and the mechanical rotation angle controller **403**, or in another location. The mechanical elevation angle controller **402** and the mechanical rotation angle controller **403** correspond to an actuator according to the embodiments. The orientation controller according to the present embodiment can control the elevation angle of the antenna **401** by controlling the mechanical elevation angle controller **402**, and can control the azimuth angle of the antenna **401** by controlling the mechanical rotation angle controller **403**. The microwave power transfer apparatus **400** is mounted on a ceiling **460**.

(61) The mechanical elevation angle controller **402** includes a mover such as a servo motor, and can use the mover to change the orientation of the antenna **401** in an elevation angle direction **D11**. With this configuration, the elevation angle of the antenna **401** can be changed. Besides a servo motor, a stepping motor or other motor may also be used as the mover. The servo motor may be of any type, such as a DC servo motor, an AC servo motor, a coreless motor, or a DD motor. Note that the case in which the orientation of the antenna **401** is controlled by the user is not ruled out. In this case, information indicating the orientation (elevation angle) to be set may be notified to the user, and the user may adjust the orientation of the antenna **401** manually. In this case, the mechanical elevation angle controller **402** may also be a mechanism supporting the antenna **401** so as to allow the orientation of the antenna **401** to be changed according to manual manipulation by the user. To notify information to the user, a communicator that communicates with a user terminal may be provided to the microwave power transfer apparatus **400**, or a display that displays information may be provided to the microwave power transfer apparatus **400**.

(62) The mechanical rotation angle controller **403** includes a mover such as a stepping motor, and can rotate itself to change the orientation of the antenna **401** in an azimuth angle direction (rotation angle direction) **D12**. With this configuration, the azimuth angle of the antenna **401** can be changed. The mechanical rotation angle controller **403** includes a support **440** supporting the antenna **401** or the mechanical elevation angle controller **402**, and can rotate the support to change the orientation of the antenna **401** in the azimuth angle direction **D12**. Besides a stepping motor, a servo motor or the like may also be used as the mover. The type of stepping motor may be any of the VR, PM, and HB types. Note that the case in which the azimuth angle of the antenna **401** is controlled by the user is not ruled out. In this case, information indicating the orientation (azimuth angle) to be set may be notified to the user, and the user may adjust the azimuth angle of the antenna **401** manually. In this case, mechanical rotation angle controller **403** may also be a mechanism supporting the support **440** so as to allow rotation according to manual manipulation by the user. To notify information to the user, a communicator that communicates with a user terminal may be provided to the microwave power transfer apparatus **400**, or a display that displays information may be provided to the microwave power transfer apparatus **400**.

(63) As above, according to the present embodiment, by controlling the azimuth angle of the antenna **401** with the mechanical rotation angle controller **403**, radiation in any azimuth angle direction is possible. Moreover, control of the plane of polarization (linear polarization) of the radiated signal is also possible through a combination of an elevation angle and an azimuth angle.

Fifth Embodiment

(64) FIG. **10** is a diagram illustrating a configuration example of a microwave power transfer apparatus **500** as an electronic apparatus according to a fifth embodiment.

(65) The microwave power transfer apparatus **500** is provided with an antenna **501**, a mechanical elevation angle controller **502**, and a mechanical rotation angle controller **503**. The antenna **501** has the same function as the antenna **101** in the first embodiment. The high-frequency device and controller (orientation controller/power controller) according to any of the first to third embodiments are provided inside the housing of at least one of the mechanical elevation angle controller **502** and the mechanical rotation angle controller **503**, or in another location. The mechanical elevation angle controller **502** and the mechanical rotation angle controller **503** correspond to an actuator according to the embodiments. The orientation controller according to the present embodiment can control the elevation angle of the antenna **501** by controlling the mechanical elevation angle controller **502**, and can control the azimuth angle of the antenna **501** by controlling the mechanical rotation angle controller **503**.

(66) The mechanical elevation angle controller **502** is provided with a mechanism that moves the antenna **501** on an arc-shaped rail **504**. The arc-shaped rail **504** is an example of a concave rail, and may also be a rail that is curved or another shape. By controlling the mechanism, the mechanical elevation angle controller **502** can change the elevation angle of the antenna **501** in an elevation angle direction **D21**. The mechanical rotation angle controller **503** is similar to the mechanical rotation angle controller of the fourth embodiment. The mechanical rotation angle controller **503** can change the orientation of the antenna **501** in an azimuth angle direction **D22**.

(67) As above, according to the present embodiment, by moving the antenna **501** on the arc-shaped rail **504** to change the elevation angle of the antenna **501**, the elevation angle is changed easily, even if the structure of the antenna makes it difficult to control the elevation angle of the antenna with the configuration in FIG. **9** described above. Moreover, by controlling the azimuth angle of the antenna **501** with the mechanical rotation angle controller **503**, radiation in any azimuth angle direction is possible. Moreover, control of the plane of polarization (linear polarization) of the radiated signal is also possible through a combination of an elevation angle and an azimuth angle.

Sixth Embodiment

(68) FIG. **11** is a diagram illustrating a configuration example of a microwave power transfer apparatus **600** as an electronic apparatus according to a sixth embodiment.

(69) The microwave power transfer apparatus **600** is provided with an antenna **601**, a mechanical elevation angle controller **602**, and a mechanical rotation angle controller **603**. The antenna **601** has the same function as the antenna **101** in the first embodiment. The high-frequency device and controller (orientation controller/power controller) according to any of the first to third embodiments are provided inside the housing of at least one of the mechanical elevation angle controller **602** and the mechanical rotation angle controller **603**, or in another location. The mechanical elevation angle controller **602** and the mechanical rotation angle controller **603** correspond to an actuator according to the embodiments. The orientation controller according to the present embodiment can control the elevation angle of the antenna **601** by controlling the mechanical elevation angle controller **602**, and can control the azimuth angle of the antenna **601** by controlling the mechanical rotation angle controller **603**.

(70) The mechanical elevation angle controller **602** includes a robot arm, and can freely change the elevation angle of the antenna **601** through the driving of the robot arm. One end of the robot arm is connected to a base **604**, and the other end of the robot arm is connected to the mechanical rotation angle controller **603**.

(71) As above, according to the present embodiment, by using the robot arm, the elevation angle of the antenna **601** can be changed in an elevation angle direction **D31**, and the antenna **601** can be moved over a wide range. Thus, the area treated as the target of power transfer can be increased. Moreover, by controlling the azimuth angle of the antenna **601** with the mechanical rotation angle

controller **603**, radiation in any azimuth angle direction D32 is possible. Moreover, control of the plane of polarization (linear polarization) of the radiated signal is also possible through a combination of an elevation angle and an azimuth angle.

Seventh Embodiment

(72) FIG. **12** is a diagram illustrating a configuration example of a microwave power transfer apparatus **700** as an electronic apparatus according to a seventh embodiment. The microwave power transfer apparatus **700** is mounted on a ceiling **760**.

(73) The microwave power transfer apparatus **700** is provided with an antenna **701** (**701A**, **701B**), a mechanical elevation angle controller **702** (actuator), and a transmitter **710**. The transmitter **710** includes the high-frequency device and controller (orientation controller/power controller) according to any of the first to third embodiments.

(74) The antenna **701** includes a radiator (antenna element) **701A** that radiates electromagnetic waves and a reflector **701B** that faces the radiator **701A** and reflects electromagnetic waves radiated from the radiator **701A**.

(75) The mechanical elevation angle controller **702** can move the reflector **701B** relative to the radiator **701A**. For example, the reflector **701B** is movable in a left circumferential direction D41 and a right circumferential direction D42. By changing the orientation of the reflector **701B** relative to the radiator **701A**, it is possible to change the reflection direction of electromagnetic waves radiated from the radiator **701A**, that is, the maximum gain direction of the antenna **701** or the radiation direction of the antenna **701**. The orientation controller according to the present embodiment moves the reflector **701B** by controlling the mechanical elevation angle controller **702**. With this configuration, an effect similar to changing the elevation angle of the antenna in the first to sixth embodiments can be obtained. Moving the reflector **701B** in this way corresponds to changing the elevation angle in the first to sixth embodiments.

(76) A mechanical rotation angle controller may also be provided as in FIGS. **9** to **11** described above, so that the radiator **701A** of the antenna **701** is also rotatable in the azimuth angle direction.

(77) As above, according to the present embodiment, even if the structure of the antenna makes it difficult to change the elevation angle, by using the antenna as the radiator **701A** and using the reflector **701B** to reflect electromagnetic waves radiated by the radiator **701A**, an effect similar to changing the elevation angle of the antenna can be obtained.

Eighth Embodiment

(78) FIG. **13** is a diagram illustrating a configuration example of a microwave power transfer apparatus **800** as an electronic apparatus according to an eighth embodiment. The microwave power transfer apparatus **800** is mounted on a ceiling **860**.

(79) The microwave power transfer apparatus **800** is provided with an antenna **801**, a mechanical elevation angle controller **802**, a mechanical rotation angle controller **803**, and an electromagnetic wave absorption/reflection plate **804**. The antenna **801** has the same function as the antenna **101** in the first embodiment. The high-frequency device and controller (orientation controller/power controller) according to any of the first to third embodiments are provided inside the housing of at least one of the mechanical elevation angle controller **802** and the mechanical rotation angle controller **803**, or in another location. The mechanical elevation angle controller **802** and the mechanical rotation angle controller **803** correspond to an actuator according to the embodiments. The orientation controller according to the present embodiment can control the elevation angle of the antenna **801** by controlling the mechanical elevation angle controller **802**, and can control the azimuth angle of the antenna **801** by controlling the mechanical rotation angle controller **803**. The configuration in FIG. **13** corresponds to the configuration in FIG. **9** with the addition of the electromagnetic wave absorption/reflection plate **804**. Hereinafter, the description that is the same as FIG. **9** is omitted, as appropriate.

(80) The electromagnetic wave absorption/reflection plate **804** is mounted on the ceiling **860** so as to enclose the antenna **801** from around the antenna **801**. In the diagram, a cross section of the

electromagnetic wave absorption/reflection plate **804** is illustrated. The electromagnetic wave absorption/reflection plate **804** absorbs or reflects electromagnetic waves radiated in the horizontal direction (90 degrees or 270 (−90 degrees)) or a neighboring direction.

(81) In the reflection case, the electromagnetic wave absorption/reflection plate **804**, which serves as a second member configured to reflect at least a portion of electromagnetic waves radiated from the antenna **801**, is provided in a location including a first location H1 corresponding to a first direction (for example, the horizontal direction) that blocks leakage of the electromagnetic waves and a second location H2 opposite the first location H1. Reflected electromagnetic waves are re-reflected by the plate on the opposite side, and are repeatedly reflected thereafter until the electromagnetic waves attenuate. Alternatively, the electromagnetic waves are converted into thermal energy and attenuated by the electromagnetic wave absorption/reflection plate **804**. With this arrangement, the leakage of electromagnetic waves in the horizontal direction can be mitigated.

(82) In the absorption case, the electromagnetic wave absorption/reflection plate **804**, which serves as a first member configured to absorb at least a portion of electromagnetic waves radiated from the antenna **801**, is provided in the location H1 corresponding to a first direction (for example, the horizontal direction) that blocks leakage of the electromagnetic waves, in the second location H2, or both.

(83) In the case of absorbing electromagnetic waves, the electromagnetic wave absorption/reflection plate **804** may be formed from a dielectric or magnetic body, for example, whereas in the case of reflecting electromagnetic waves, the electromagnetic wave absorption/reflection plate **804** may be formed from a metal. This configuration makes it possible to mitigate interference on another wireless system existing in the horizontal direction or a neighboring direction, regardless of the elevation angle of the antenna **801**.

(84) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

(85) The embodiments as described before may be configured as below.

CLAUSES

(86) Clause 1. An electronic apparatus capable of transferring power wirelessly to a power receiver apparatus, apparatus comprising: a signal generation circuitry configured to generate a first signal for wireless power transfer; an antenna configured to radiate an electromagnetic wave on a basis of the first signal; an actuator capable of mechanically changing an orientation of the antenna; and a controlling circuitry configured to control a transmission power of the first signal according to the orientation of the antenna. Clause 2. The electronic apparatus according to clause 1, wherein the signal generation circuitry includes at least one of a variable amplifier capable of changing a gain of the first signal and a variable attenuator capable of changing an attenuation of the first signal, and the controlling circuitry is configured to control the transmission power of the first signal by controlling at least one of the variable amplifier and the variable attenuator. Clause 3. The electronic apparatus according to clause 1 or 2, wherein the signal generation circuitry includes an amplifier and a variable attenuator, and the controlling circuitry is configured to control the transmission power of the first signal by controlling the variable attenuator. Clause 4. The electronic apparatus according to clause 2 or 3, wherein the controlling circuitry is configured to control the transmission power of the first signal on a basis of correspondence data associating a plurality of orientations that the antenna may take with a plurality of attenuation values or a plurality of gain values. Clause 5. The electronic apparatus according to any one of clauses 1 to 4, wherein the actuator is capable of changing at least an elevation angle from among an elevation

angle and an azimuth angle of the antenna. Clause 6. The electronic apparatus according to clause 5, wherein the controlling circuitry is configured to control the transmission power of the first signal to a first power value in a case in which the elevation angle of the antenna points in a first direction, and control the transmission power of the first signal to a second power value smaller than the first power value in a case in which the elevation angle of the antenna points in a second direction closer to a direction horizontal to an installation surface of the electronic apparatus than the first direction. Clause 7. The electronic apparatus according to any one of clauses 1 to 6, further comprising: a receive processing circuitry configured to perform, through the antenna, carrier sense of a second signal from another wireless system different from the power receiver apparatus, wherein the controlling circuitry is configured to control the actuator on a basis of a result of the carrier sense. Clause 8. The electronic apparatus according to clause 7, wherein the receive processing circuitry is configured to compare a receive power level of the second signal to a threshold value as the carrier sense, and the controlling circuitry is configured to change the threshold value according to the orientation of the antenna. Clause 9. The electronic apparatus according to clause 8, wherein when the receive power level is greater than the threshold value, the controlling circuitry is configured to control the actuator to change the orientation of the antenna, or to suspend transmission of the electromagnetic waves from the antenna. Clause 10. The electronic apparatus according to clause 8 or 9, wherein the receive processing circuitry is configured to use a first threshold value in a case in which the elevation angle of the antenna points in a first direction, and use a second threshold value larger than the first threshold value in a case in which the elevation angle of the antenna points in a second direction closer to a direction horizontal to an installation surface of the electronic apparatus than the first direction. Clause 11. The electronic apparatus according to clause 5 or 6, wherein the actuator is capable of changing the elevation angle of the antenna by moving the antenna along a concave rail. Clause 12. The electronic apparatus according to clause 5, 6 or 11, wherein the actuator includes a robot arm having one end connected to a base, the antenna is connected to another end of the robot arm, and the actuator is capable of changing the elevation angle of the antenna by driving the robot arm. Clause 13. The electronic apparatus according to clause 5, 6, 11 or 12 wherein the actuator is capable of changing the azimuth angle of the antenna by rotating a support supporting the antenna. Clause 14. The electronic apparatus according to any one of clauses 1 to 13, wherein the antenna is provided with a radiator configured to radiate the electromagnetic waves and a reflector configured to reflect the electromagnetic waves radiated from the radiator, and the antenna is capable of changing a direction in which the electromagnetic waves are reflected by moving the reflector relative to the radiator. Clause 15. The electronic apparatus according to any one of clauses 1 to 14, wherein a first member capable of absorbing at least a portion of electromagnetic waves radiated from the antenna is provided in a location corresponding to a first direction that blocks leakage of the electromagnetic waves. Clause 16. The electronic apparatus according to any one of clauses 1 to 15, wherein a second member capable of reflecting at least a portion of electromagnetic waves radiated from the antenna is provided in a location including a first location corresponding to a first direction that blocks leakage of the electromagnetic waves and a second location opposite the first location. Clause 17. The electronic apparatus according to any one of clauses 1 to 16, wherein the controlling circuitry is configured to control the transmission power of the first signal on a basis of maximum output characteristics stipulating a maximum output value that the antenna may take in each elevation angle direction. Clause 18. The electronic apparatus according to clause 17, wherein the maximum output value of a radiation level of the electromagnetic waves in a direction horizontal to an installation surface of the electronic apparatus is less than or equal to 1/100 a maximum radiation level of the antenna. Clause 19. A wireless power transfer method of transferring power wirelessly to a power receiver apparatus, comprising: mechanically changing an orientation of an antenna relative to the power receiver apparatus; determining a transmission power of a first signal for the wireless power transfer according to the orientation of the antenna;

generating the first signal on a basis of the transmission power; and radiating electromagnetic waves from the antenna on a basis of the first signal. Clause 20. A wireless power transfer system comprising: a power receiver apparatus; a signal generation circuitry configured to generate a first signal for wireless power transfer to the power receiver apparatus; an antenna configured to radiate electromagnetic waves on a basis of the first signal; an actuator capable of mechanically changing an orientation of the antenna; and a power control circuit configured to control a transmission power of the first signal according to the orientation of the antenna.

Claims

1. An electronic apparatus capable of transferring power wirelessly to a power receiver apparatus, apparatus comprising: a signal generation circuitry configured to generate a first signal for wireless power transfer; an antenna configured to radiate an electromagnetic wave; an actuator capable of mechanically changing an orientation of the antenna, the orientation being changeable at least between a first orientation and a second orientation, the first orientation corresponding to a direction horizontal to an installation surface of the electronic apparatus and the second orientation corresponding to a direction vertical to the installation surface of the electronic apparatus; at least one of a variable amplifier capable of changing a gain of the first signal and a variable attenuator capable of changing an attenuation of the first signal; a memory configured to store correspondence data associating a plurality of orientations that the antenna may take with a plurality of attenuation values or a plurality of gain values, the plurality of orientations including the first orientation and the second orientation, and the correspondence data being configured such that the transmission power increases as the orientation of the antenna becomes closer from the first orientation to the second orientation; and a controlling circuitry configured to control the transmission power of the electromagnetic wave according to the orientation of the antenna by controlling at least one of the variable amplifier or the variable attenuator on the basis of the correspondence data.
2. The electronic apparatus according to claim 1, wherein the signal generation circuitry includes an amplifier and a variable attenuator, and the controlling circuitry is configured to control the transmission power of the electromagnetic wave by controlling the variable attenuator.
3. The electronic apparatus according to claim 1, wherein the actuator is capable of changing at least an elevation angle from among an elevation angle and an azimuth angle of the antenna.
4. The electronic apparatus according to claim 3, wherein the controlling circuitry is configured to control the transmission power of the first signal to a first power value in a case in which the elevation angle of the antenna points in a first direction, and control the transmission power of the first signal to a second power value smaller than the first power value in a case in which the elevation angle of the antenna points in a second direction closer to a direction horizontal to an installation surface of the electronic apparatus than the first direction.
5. The electronic apparatus according to claim 3, wherein the actuator is capable of changing the elevation angle of the antenna by moving the antenna along a concave rail.
6. The electronic apparatus according to claim 3, wherein the actuator includes a robot arm having one end connected to a base, the antenna is connected to another end of the robot arm, and the actuator is capable of changing the elevation angle of the antenna by driving the robot arm.
7. The electronic apparatus according to claim 3, wherein the actuator is capable of changing the azimuth angle of the antenna by rotating a support supporting the antenna.
8. The electronic apparatus according to claim 1, further comprising: a receive processing circuitry configured to perform, through the antenna, carrier sense of a second signal from another wireless system different from the power receiver apparatus, wherein the controlling circuitry is configured to control the actuator on a basis of a result of the carrier sense.
9. The electronic apparatus according to claim 8, wherein the receive processing circuitry is configured to compare a receive power level of the second signal to a threshold value as the carrier

sense, and the controlling circuitry is configured to change the threshold value according to the orientation of the antenna.

10. The electronic apparatus according to claim 9, wherein when the receive power level is greater than the threshold value, the controlling circuitry is configured to control the actuator to change the orientation of the antenna, or to suspend transmission of the electromagnetic waves from the antenna.

11. The electronic apparatus according to claim 9, wherein the receive processing circuitry is configured to use a first threshold value in a case in which the elevation angle of the antenna points in a first direction, and use a second threshold value larger than the first threshold value in a case in which the elevation angle of the antenna points in a second direction closer to a direction horizontal to an installation surface of the electronic apparatus than the first direction.

12. The electronic apparatus according to claim 1, wherein the antenna is provided with a radiator configured to radiate the electromagnetic waves and a reflector configured to reflect the electromagnetic waves radiated from the radiator, and the antenna is capable of changing a direction in which the electromagnetic waves are reflected by moving the reflector relative to the radiator.

13. The electronic apparatus according to claim 1, wherein a first member capable of absorbing at least a portion of electromagnetic waves radiated from the antenna is provided in a location corresponding to a first direction that blocks leakage of the electromagnetic waves.

14. The electronic apparatus according to claim 1, wherein a second member capable of reflecting at least a portion of electromagnetic waves radiated from the antenna is provided in a location including a first location corresponding to a first direction that blocks leakage of the electromagnetic waves and a second location opposite the first location.

15. The electronic apparatus according to claim 1, wherein the controlling circuitry is configured to control the transmission power of the first signal on a basis of maximum output characteristics stipulating a maximum output value that the antenna may take in each elevation angle direction.

16. The electronic apparatus according to claim 15, wherein the maximum output value of a radiation level of the electromagnetic waves in a direction horizontal to an installation surface of the electronic apparatus is less than or equal to $1/100$ a maximum radiation level of the antenna.

17. A wireless power transfer method of transferring power wirelessly to a power receiver apparatus, comprising: mechanically changing an orientation of an antenna relative to the power receiver apparatus, the orientation being changeable at least between a first orientation and a second orientation, the first orientation corresponding to a direction horizontal to an installation surface of a power transfer apparatus and the second orientation corresponding to a direction vertical to the installation surface of the power transfer apparatus; determining a transmission power of an electromagnetic wave for the wireless power transfer according to the orientation of the antenna; generating a first signal on a basis of the transmission power; radiating the electromagnetic wave from the antenna; controlling the transmission power of the electromagnetic wave according to the orientation of the antenna by controlling at least one of a variable amplifier capable of changing a gain of the first signal and a variable attenuator capable of changing an attenuation of the first signal; storing, in a memory, correspondence data associating a plurality of orientations that the antenna may take with a plurality of attenuation values or a plurality of gain values, and the correspondence data being configured such that the transmission power increases as the orientation of the antenna becomes closer from the first orientation to the second orientation; and controlling the transmission power of the on a basis of the correspondence data.
